Laboratory Predictions - 30-Day Workload Forecast

Plain-English Summary of the predictions notebook

What this report is

- Forecasts the laboratory's workload for the next 30 days from one full year of order data (72,948 test orders, Jan 2025 - Jan 2026).
- Two forecast levels are shown as charts below - workload per department and pressure per analyzer model - with all numbers in plain units: tests per day, run minutes and busy % of a working day.

How the forecast works (in plain words)

- One simple, explainable model: it learns how demand moves with the day of the week and the month.
- Tested honestly: trained on ~11 months, then asked to predict the final 30 days it had never seen.
- Forecasts target typical days - spike days (e.g. 1 Jan 2026, 5,366 orders in one day) are documented separately and land on top of any forecast.

How accurate is it? (the 30-day holdout test)

- Departments: within ~2-5 orders/day on typical days (Serology, the busiest: 4.8).
- Analyzers (the 10 busiest models): within 1.5-2.9 tests/day.
- It roughly matches the 'use the average' baseline - demand is so regular that the calendar itself explains it; nothing fancier is justified.
- With spike days included the error looks far larger (up to 31 orders/day) - that is the spikes, not the model.

Key numbers at a glance

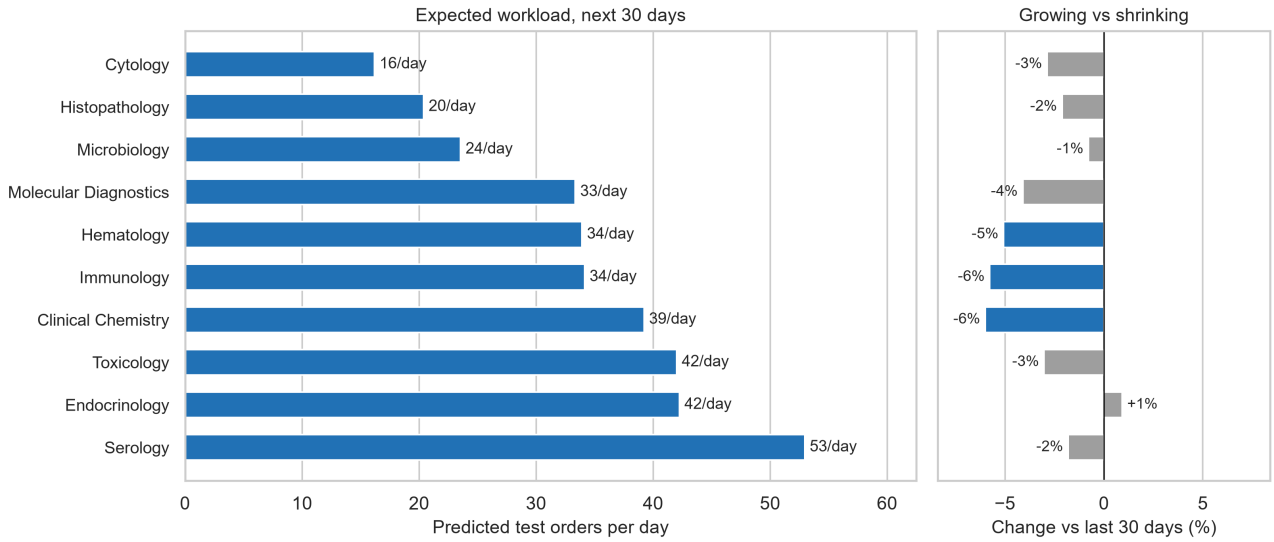
- Departments: Serology stays heaviest (~53 orders/day, ~1,589 in 30 days); the lab expects ~10,140 orders, ~3% below the last 30 days; no department moves beyond about 6% either way.
- Analyzers: the busiest model uses ~11% of one 2-shift day (14 h) per machine, the median is 0.8%, and all 50 models stay in 'plenty of spare time' - no machine purchase is triggered.

Honesty notes and limits

- Typical-day forecasts only - an unpredicted spike-type event hits the slow-throughput machines hardest.
- The 14 h two-shift lab day is an assumption inherited from the operational report (§11); it rescales the busy % but does not change rankings or verdicts.

Department workload - forecast for the next 30 days

Predicted test orders per day for each department (left) and how each department differs from the last 30 days (right). Red means growing, blue means shrinking, grey means steady.



Key insights

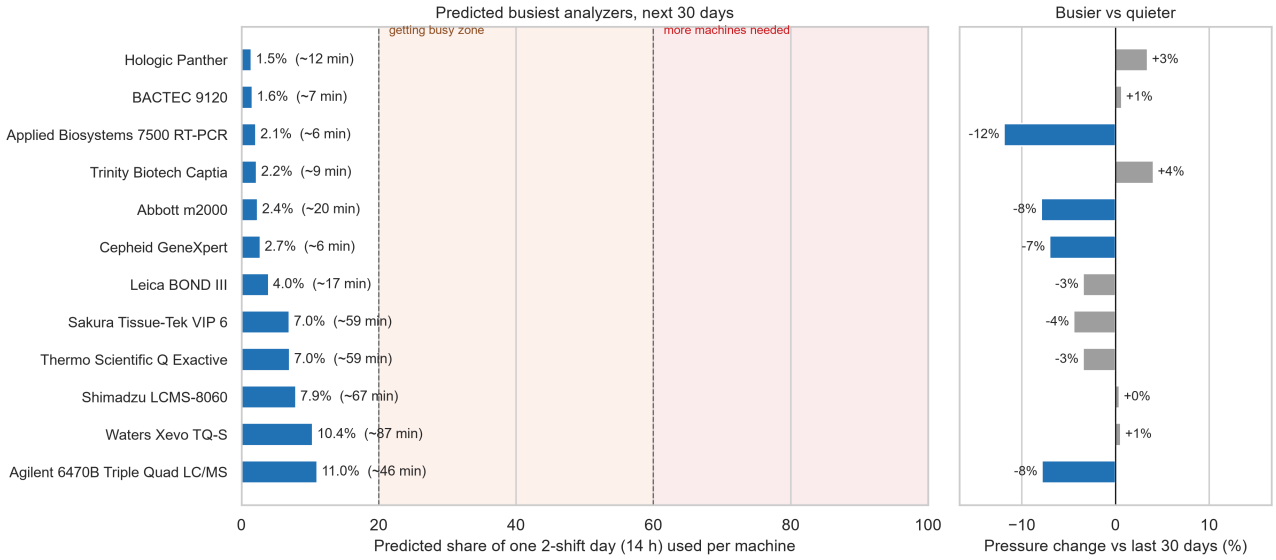
- Serology remains the heaviest department: ~53 orders/day (~1,589 in 30 days) - plan its rosters first.
- Endocrinology (~42/day) and Toxicology (~42/day) follow; Cytology is the lightest at ~16/day.
- Lab-wide the forecast expects ~10,140 test orders, ~3% below the last 30 days - a steady picture.
- Biggest department move: Clinical Chemistry (-6%, about 2.5 fewer orders/day); smallest: Microbiology (-0.8%). No department surges or collapses.

What this means for planning

- Roster the top three departments (Serology, Endocrinology, Toxicology) as for a normal month - no surge staffing is indicated anywhere.
- If a spike-type event recurs (like 1 Jan 2026), handle it with a separate emergency protocol - it arrives on top of these numbers.

Analyzer workload pressure - forecast for the next 30 days

The 12 busiest analyzer models as a share of one 2-shift day (14 h) per machine (left; shaded bands mark the 'getting busy' and 'more machines' zones) and the pressure change vs the last 30 days (right).



Key insights

- The busiest model, the Agilent 6470B Triple Quad LC/MS (Toxicology), uses only ~11% of one 2-shift day per machine (~46 run minutes/day) - deep inside 'plenty of spare time'.
- Waters Xevo TQ-S tops run-minutes (~87 min per machine per day): only ~9 tests a day, but each takes long at 6 tests/hour.
- Pressure follows throughput, not order volume: fast chemistry analysers (250-300 tests/hour) barely register (~0.1-0.2% busy) despite similar order counts.
- All 50 analyzer models stay in 'plenty of spare time' (median 0.8% busy), and the picture barely moves: the busiest dozen changes within about 8% vs the last 30 days (Applied Biosystems 7500 RT-PCR eases the most, -12%).

What this means for planning

- No new machines are needed next month - the forecast confirms the operational report's verdict.
- Monitor the slow LC-MS / toxicology machines first if volume rises: each extra test there buys run time at a fixed, slow rate.
- Idle capacity (especially the fast chemistry fleet) is available for QC runs, re-runs and backlog catch-up.